

Baseline Deepfake Detection Experiments Report

Objective

The goal of this stage of the project was to:

- Establish a working deepfake detection pipeline
- Train and evaluate two different backbone architectures:
 - MobileNetV2 (CNN-based)
 - Vision Transformer (ViT-based)
- Perform video-level evaluation on FaceForensics++ C23
- Validate dataset handling, augmentation, aggregation, and metric computation

This stage focuses on pipeline correctness and reproducibility, not final peak performance.

Dataset

FaceForensics++ C23

- Dataset variant: FaceForensics++ (C23 compression level)
- Frame-based training (extracted faces)
- Video-level evaluation (aggregation of frame probabilities)

Data Split

- Train videos: 1700 folders
- Validation videos: derived split
- Test videos: 712 videos
 - Real: 164
 - Fake: 548

The dataset is highly imbalanced, with ~77% fake videos in the test split.

Labeling Strategy

Since frames were stored as “split/<video_id>/frame_*.jpg”. Labels were inferred using _id_map.csv files. Path contains /original/ → label = 0 (real). Otherwise → label = 1 (fake).

Data Processing & Augmentation

Implemented in augmentations.py:

- RandomResizedCrop (mild scaling)
- Horizontal Flip (p=0.5)
- ColorJitter (low intensity)
- RandomErasing (small region)
- ImageNet normalization

Purpose:

- Improve generalization
- Reduce overfitting to compression artifacts
- Preserve facial structure

Evaluation Transforms

- Resize
- ToTensor
- ImageNet normalization

No random augmentations used during validation or testing.

Model Architectures

- MobileNetV2 Baseline
 - Pretrained on ImageNet
 - Final classifier replaced with 1-logit binary head
 - Loss: BCEWithLogitsLoss
 - Class imbalance handled using pos_weight

Advantages:

- Lightweight
- Fast to train
- Suitable for edge deployment
- Vision Transformer (ViT)
 - Pretrained ViT backbone
 - Linear binary classification head
 - Same training protocol (loss, optimizer, aggregation)

Advantages:

- Global attention mechanism
- Better modeling of spatial inconsistencies
- Stronger representation learning potential

Training Setup

- Optimizer: AdamW
- Learning rate: 5e-5
- Epochs: 20
- Best model saved using validation loss
- Device: CPU (no CUDA available)

Video-level prediction computed as “video_score = mean(frame_probabilities)” =>

Threshold = 0.5

Evaluation Metrics

- Accuracy
- ROC-AUC
- Average Precision (AP)
- Confusion Matrix
- Precision / Recall / F1-score

Experimental Results (MobileNetV2 Baseline)

Test ser: 712 videos

Metric	Value
Accuracy	0.4087
ROC-AUC	0.4496
Average Precision	0.7402

Confusion Matrix:

[[100 64]

[357 191]]

Observations:

- Real recall improved in some runs (~0.60)
- Fake recall degraded (~0.35)
- ROC-AUC < 0.5 suggests weak ranking capability
- High AP influenced by class imbalance (fake majority)

Experimental Insights

1. Pipeline is fully functional
 - Data loading via mapping files works
 - Augmentations apply correctly
 - Video-level aggregation works
 - Metrics are correctly computed and saved
2. Dataset imbalance is significant
 - 548 fake vs 164 real in test
 - Naive "always fake" baseline $\approx$ 77% accuracy
 - Accuracy alone is misleading
 - ROC-AUC is a better performance indicator
3. Early overfitting observed

- Best validation loss occurred early (epoch 1–2)
 - Later epochs degraded validation performance
 - Suggests:
 - Limited generalization
 - High inter-frame similarity
 - Dataset compression artifacts
4. Current baseline is a pipeline validation stage
- End-to-end training works
 - Evaluation protocol works
 - Architecture switching (MobileNet ↔ ViT) works

ViT vs MobileNet (Conceptual Comparison)

Aspect	MobileNetV2	ViT
Architecture	CNN	Transformer
Inductive bias	Strong spatial locality	Global attention
Parameter efficiency	Lightweight	Heavier
Edge deployment	Suitable	Less suitable
Representation power	Moderate	High

Conclusion of Baseline Stage

This phase successfully established:

- A modular deepfake detection framework
- Backbone interchangeability
- Frame-to-video aggregation strategy
- Class imbalance handling
- Reproducible experiment logging

While performance is not yet optimal, the pipeline is stable and ready for:

- Multi-dataset training
- Balanced sampling strategies
- Advanced temporal modeling
- Threshold tuning
- Feature-level fusion approaches